

Solution Architecture

Pharma R&D Convergence Hub (ARCH)

Aligning Knowledge Integration with Pharmaceutical R&D Excellence
24-Month Strategic Implementation Roadmap

Enterprise Knowledge Accelerator (EKA)
Knowledge Integration & Articulation (KIA)

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Executive Summary

This Solution Architecture defines the technical blueprint and 24-month implementation roadmap for the Pharma R&D Convergence Hub (ARCH), a unified knowledge platform engineered to transform Pharma's research enterprise from a data-driven to a knowledge-driven paradigm. ARCH integrates a harmonized 124TB data corpus, a labeled property graph containing over 30 million nodes and nearly 2 billion relationships, and a federated AI layer to deliver traceable, hallucination-free insight at the point of scientific decision-making.

The platform addresses a competitive mandate: as Pharma expands its footprint in high-complexity therapeutic areas—Oncology, Immunology, and Neuroscience—the ability to validate a target's causal role in disease determines the difference between a multi-billion-dollar success and a clinical failure. ARCH compresses discovery timelines from months to minutes, reduces late-stage attrition, and delivers quantified business value across the entire drug development lifecycle.

Strategic Pillars

- **Knowledge Democratization:** Serving as the trusted knowledge partner for R&D and Corporate Strategy, ensuring decision-making is fueled by shared intelligence rather than fragmented silos.
- **AI-Enabled Insight:** Powering enterprise AI platforms by integrating published external literature with formalized internal insights to drive high-velocity discovery.
- **Information Unlocking:** Leveraging patented NLP and machine learning to extract actionable relationships from a harmonized global corpus, unlocking information that makes cures possible.

Document Scope

This document serves as the authoritative reference for the ARCH solution architecture. It encompasses the platform's layered technical architecture, a phased 24-month implementation roadmap with explicit deliverables and success criteria, advanced clinical capabilities including Patient Digital Twins and Digital Health Technology integration, data governance standards anchored in FAIR Principles and MBSE methodology, specialized user experience platforms, semantic governance frameworks, and strategic risk mitigation strategies.

Layered Solution Architecture

The ARCH platform is engineered as a four-layer convergence architecture, each layer building upon the previous to transform raw data into actionable, traceable knowledge. This “Architecture of Truth” utilizes a bi-layered data and knowledge approach, breaking the organizational silos that traditionally separate medical affairs, early discovery, and clinical research.

Layer 1: Data Foundation & Ingestion

The foundational layer manages the harmonized data corpus—a massive 124TB environment connecting 200+ internal and external data sources. This layer is governed by strict FAIR Principles (Findability, Accessibility, Interoperability, Reusability) and enforces unyielding Service Level Agreements (SLAs) for all ingested data assets.

Core Technical Components

- **Data Movement & Automation (Modak Nabu):** Manages underlying data flow, automated ingestion pipelines, and ETL orchestration at enterprise scale. All ingestion workflows are designed using Model-Based Systems Engineering (MBSE) principles to guarantee zero data loss during integration from legacy systems and disparate source schemas.
- **Named Entity Recognition (SciBite TERMite):** Normalizes biomedical concepts—genes, proteins, chemicals, diseases—with high precision, achieving “PhD accuracy at Big Data scale” through synonym mapping to a single ontological concept.
- **Ontology Management (SciBite CENTree):** Democratizes the creation and management of enterprise-wide controlled vocabularies, providing the semantic backbone that transforms disparate strings into unified biological things.
- **FAIR Data Quality Gateway:** Every data asset entering ARCH must pass through a quality gateway that validates compliance with FAIR Principles. Automated quality scoring assigns Findability, Accessibility, Interoperability, and Reusability metrics before data is promoted to the knowledge layer.

Layer 2: Semantic Harmonization & Knowledge Graph

This layer transforms normalized data into structured, semantically enriched knowledge. The Pharma Repository of Knowledge (ARK) is a labeled property graph managing over 30 million nodes and nearly 2 billion relationships, providing a comprehensive map of biomedical cause-and-effect.

Core Technical Components

- **Relationship Extraction (Tellic):** Extracts over 1 billion biomedical relationships from the global literature corpus, transitioning from statistical co-occurrence to relationship-theme extraction using distributional semantics. The ARK graph is populated with 80+ distinct relationship types (e.g., inhibits, agonism, upregulates) to model complex biological interactions.
- **Neo4j Semantic Engine:** Serves as the graph database engine, enabling Cypher-based querying of the knowledge graph with full provenance tracing. Every triple is tagged with source, timestamp, and confidence metadata.

- **Fidelity Threshold Enforcement:** A high threshold value (0.9) is applied to all NLP-derived triples. Insights from the GNBR-Nexus evaluation confirm that applying high thresholds to specific relationship classes like “inhibits” yields high-fidelity triples not reported in structured datasets, ensuring the Knowledge Graph remains an Architecture of Truth.
- **Entity Resolution & the Reveal-Reframe-Resist Model:** A governance layer that identifies NLP failures (e.g., confusing COX-1 abbreviations), ensures context-driven mapping to precise concept IDs in ontologies like UniProt, and uses expert curation combined with LLM grounding to resist the incorporation of mislabeled entities.

Layer 3: AI & Prompt Engineering (Semantic Gateway)

The AI layer operationalizes the Context Engine, moving beyond simple keyword matching to deliver relevancy grounded in ontological metadata. This layer acts as the Semantic Gateway between the user and the knowledge graph, ensuring every AI interaction is contextually precise.

Core Technical Components

- **AWS Bedrock Router:** Routes user queries to the appropriate language model based on persona, permissions, and therapeutic domain context. Identifies whether the user is a Scientist, Practitioner, or Patient and applies the corresponding language model and permission set.
- **Enterprise Prompt Library:** A curated library of ontology-enriched prompt templates that wraps user requests in structured metadata—the 5 Ws (Who, What, Where, When, Why)—before the query reaches the LLM. This ensures intent-driven prompting rather than raw prompt submission.
- **Federated RAG 2.0 Architecture:** Utilizes domain-specific vector databases as virtual Specialized Language Models (SLMs), querying the Neo4j graph directly to ground LLM outputs in verified semantic data. Cypher query generation eliminates hallucinations and provides traceable provenance for every answer.
- **Context Engine (5 Pillars of Relevancy):** Maps every query through five ontological dimensions: Who (Party/Persona), What (Product/Asset), Where (Location/Domain), When (Time/Event), and Why (Issue/Service/Intent). This ensures that identical queries from different personas return contextually appropriate results.

Layer 4: User Interface & Application Delivery

The presentation layer delivers intuitive, task-tailored applications for every stakeholder persona through the Unity design system (TXI Digital). This layer provides 18 distinct views of a molecule or gene target, reducing time-to-insight from months to minutes.

Platform Portfolio

- **ARCH Search:** The primary search interface enabling natural-language querying of the knowledge graph, with results grounded in ARK data and full provenance tracing.
- **Target Dossier:** A comprehensive view aggregating pathways, variants, phenotypes, and safety signals for a specific biological target, replacing months of manual curation with a unified, real-time interface.

- **Safety Dossier:** Aggregates adverse event data, toxicology findings, and safety signals across internal clinical trials and external literature, rendered as an auditable, GxP-compliant view.
- **Pharma Science Oncology Platform:** A modernized, specialized interface delivering tailored research updates directly to healthcare practitioners in oncology. This platform extends the ARCH ecosystem beyond internal R&D, providing practitioners with curated, graph-grounded insights on emerging targets, clinical trial updates, and translational research relevant to their therapeutic focus.
- **Rare Hopes UX:** A purpose-built patient-and-clinician interface designed to translate complex graph-derived hypotheses into actionable insights for rare diseases. Rare Hopes transforms the dense biomedical relationships in ARK into accessible language for patients and their doctors, enabling informed shared decision-making for conditions where traditional evidence bases are sparse.
- **Ask ARCH:** A natural language conversational interface for knowledge graph querying, grounding LLM responses in structured ARK data using vector search and Cypher query generation.

Core Technology Integration Matrix

The following matrix maps each core technology component to its functional role, strategic partner, and architectural layer within the ARCH platform.

Component	Functional Role	Strategic Partner	Architecture Layer
Data Movement & Automation	Automated ingestion, ETL orchestration, MBSE-governed data flow	Modak Nabu	Layer 1: Foundation
Ontology Management	Enterprise-wide controlled vocabularies and semantic governance	SciBite CENTree	Layer 1: Foundation
Named Entity Recognition	Biomedical concept normalization with PhD-level precision	SciBite TERMite	Layers 1–2
Relationship Extraction	1B+ biomedical relationship extraction via distributional semantics	Tellic	Layer 2: Semantic
Graph Database Engine	Labeled property graph with Cypher querying and provenance tracing	Neo4j	Layer 2: Semantic
AI Orchestration & Routing	Persona-based LLM routing, Federated RAG 2.0, prompt engineering	AWS Bedrock	Layer 3: AI
UI/UX Design System	Intuitive, task-tailored applications for scientists and practitioners	TXI Digital (Unity)	Layer 4: Interface

24-Month Strategic Implementation Roadmap

The implementation follows a “quick but not cheap” mandate, structured across three phases over a precise 24-month execution timeline. Each phase builds upon the previous, delivering incremental business value while progressively advancing the platform’s capabilities from foundational data ingestion through autonomous agentic intelligence.

Phase 1: Foundation (Months 1–6)

Objective: Establish the unified data foundation, achieve semantic normalization at scale, and deliver the initial knowledge graph with verified, high-fidelity relationships.

Stage I: Data Ingestion & Normalization

- **Primary Deliverable:** A unified data lake connecting 200+ internal and external sources with full FAIR compliance.
- **MBSE-Governed Ingestion:** All data integration workflows are designed and validated using Model-Based Systems Engineering principles, guaranteeing zero data loss during migration from legacy systems and disparate source schemas. MBSE serves as the primary methodology for structurally sound integration, preventing the accumulation of integration debt.
- **FAIR Data Quality SLAs:** Every ingested data asset must satisfy unyielding Service Level Agreements across all four FAIR dimensions. Automated quality scoring gates prevent non-compliant data from entering the ARCH ecosystem. Findability mandates require persistent identifiers and rich metadata; Accessibility requires standardized retrieval protocols; Interoperability enforces controlled vocabulary mapping; Reusability demands clear provenance and usage licensing.
- **NER & Synonym Mapping:** Deployment of SciBite TERMite for biomedical concept normalization, mapping all entities to single ontological concepts via SciBite CENTree vocabularies.
- **Clinical Data De-identification:** De-identification pipelines for data from over 2,000 clinical trials, maintaining ethical exclusions for pediatric and prison populations, enabling risk-based data democratization for secondary use.
- **Measure for Success:** Achieving PhD accuracy at Big Data scale through NER and synonym mapping; 100% FAIR compliance scoring on all promoted data assets; zero data loss validated through MBSE traceability audits.

Stage II: Knowledge Extraction & Modeling

- **Primary Deliverable:** Population of the ARK labeled property graph with semantically enriched, high-fidelity biomedical relationships.
- **Distributional Semantics Extraction:** Transition from statistical co-occurrence to relationship-theme extraction using Tellic’s NLP engine, populating the graph with 80+ distinct relationship types modeling complex biological interactions.
- **Fidelity Threshold Enforcement:** Application of a 0.9 confidence threshold for all NLP-derived triples, validated against GNBR-Nexus benchmarks to ensure only high-fidelity relationships enter the graph.

- **Entity Resolution Governance:** Deployment of the Reveal-Reframe-Resist model for entity resolution, with expert curation workflows to identify and correct NLP misclassifications before graph population.
- **Measure for Success:** ARK graph populated with 30M+ nodes and initial relationship sets; all triples above 0.9 fidelity threshold; entity resolution accuracy validated by domain scientists.

Phase 2: Acceleration (Months 6–12)

Objective: Deploy user-facing applications, launch specialized UX platforms, introduce Patient Digital Twin capabilities, and establish the AI-powered Semantic Gateway for context-aware querying.

Stage III: Interface & Application Deployment

- **Primary Deliverable:** Full deployment of ARCH Search, Target and Safety Dossiers, Pharma Science Oncology Platform, and Rare Hopes UX via the Unity design system.
- **Target & Safety Dossiers:** Delivery of 18 distinct views of a molecule or gene target, aggregating pathways, variants, phenotypes, and safety signals into a unified, GxP-compliant interface. Scientists transition from months of manual curation to minutes of graph-powered insight.
- **Pharma Science Oncology Platform:** Launch of the modernized oncology interface delivering tailored research updates, emerging target intelligence, and clinical trial findings directly to healthcare practitioners. This platform extends ARCH's value beyond internal R&D into the practitioner ecosystem, providing curated graph-grounded insights for oncology decision-making.
- **Rare Hopes UX:** Deployment of the specialized patient-and-clinician interface for rare diseases. Rare Hopes translates complex graph-derived hypotheses from ARK into accessible, actionable insights for patients and their doctors, enabling shared decision-making where traditional evidence bases are sparse. The platform surfaces potential therapeutic connections that would otherwise remain buried in research literature.
- **Measure for Success:** 18 dossier views operational; time-to-insight reduced from months to minutes; Rare Hopes UX validated with rare disease patient advisory boards; Pharma Science Oncology Platform active with practitioner user base.

Patient Digital Twin Capabilities

- **Primary Deliverable:** Initial deployment of Patient Digital Twin infrastructure leveraging observational data and historical control arms.
- **Observational Data Integration:** Integration of real-world patient data, historical clinical trial outcomes, and biomarker profiles into patient-specific computational models. These Digital Twins simulate placebo outcomes for specific patient profiles, enabling researchers to predict individual responses to therapeutic interventions before physical trials begin.
- **Clinical Trial Optimization:** Digital Twin simulations are used to reduce redundant experiments by modeling expected outcomes across patient subpopulations, optimizing clinical trial sample sizes, and identifying optimal patient stratification strategies. This capability directly reduces clinical trial costs while improving statistical power.
- **Measure for Success:** Digital Twin models validated against historical control arm data with predefined concordance thresholds; demonstrated reduction in proposed trial sample sizes for at least one active program.

Context Engine & Semantic Gateway Deployment

- **Primary Deliverable:** Operational deployment of the AWS Bedrock Router, Enterprise Prompt Library, and Context Engine with full 5 Pillars of Relevancy integration.

- **Persona-Based Query Routing:** The Bedrock Router identifies user persona (Scientist, Practitioner, Patient) and applies corresponding permissions, language models, and prompt templates automatically.
- **Intent-Driven Prompting:** The Enterprise Prompt Library wraps every user request in ontological metadata (Who, What, Where, When, Why) before it reaches the LLM, ensuring that identical queries from different personas return contextually appropriate results.
- **Cypher Query Generation:** The Semantic Gateway translates enriched prompts into precise Cypher queries against the ARK graph, grounding every LLM response in verified data with full provenance tracing.
- **Measure for Success:** Context Engine resolving polymorphic meaning correctly across all three persona types; LLM responses grounded with traceable provenance for 100% of graph-sourced answers.

Phase 3: Evolution (Months 12–24)

Objective: Advance to autonomous agentic capabilities, integrate Digital Health Technology endpoints, scale Patient Digital Twins, and deploy the Gaia Platform for continuous recursive learning.

Stage IV: Generative AI & Agentic Integration

- **Primary Deliverable:** Ask ARCH conversational interface, Large Action Models (LAMs), and the Gaia Platform for autonomous knowledge evolution.
- **Ask ARCH:** Full deployment of the natural language conversational interface for knowledge graph querying. Users interact with ARCH through natural language; the system generates Cypher queries, retrieves grounded results, and articulates answers with full provenance and confidence scoring.
- **Large Action Models (LAMs):** Evolution from retrieval-only to action-oriented AI agents. LAMs autonomously execute multi-step R&D workflows—from hypothesis generation through literature validation to experimental design recommendation—mapping action model workflows to specific R&D milestones.
- **Gaia Platform:** The recursive learning platform where the ontology does not just inform but executes. Gaia enables continuous, autonomous enrichment of the knowledge graph through feedback loops from experimental outcomes, clinical trial results, and real-world evidence. The graph evolves as new knowledge is generated, maintaining its status as a living Architecture of Truth.
- **Measure for Success:** Ask ARCH handling complex multi-hop queries with >95% provenance accuracy; LAMs autonomously completing defined R&D workflow sequences; Gaia Platform demonstrating measurable graph enrichment from recursive learning cycles.

Digital Health Technology & Wearables Integration

- **Primary Deliverable:** Integration of Digital Health Technology (DHT) endpoints, including wearable device data streams, into the ARCH knowledge ecosystem.
- **Wearable Data Ingestion:** Integration of continuous physiological data from wearable devices (smartwatches, biosensors, activity trackers) into the ARCH data foundation. DHT endpoints capture objective, real-world patient outcomes—including activity levels, heart rate variability, sleep patterns, and movement biomarkers—providing continuous longitudinal data that complements traditional clinical endpoints.
- **Patient Digital Twin Maturation:** Wearable data streams feed directly into Patient Digital Twin models, enriching simulations with real-time physiological signals. This enables dynamic, continuously updated patient models that evolve beyond static historical data, providing researchers with living computational representations of patient health trajectories.
- **Real-World Evidence Generation:** DHT integration supports post-market surveillance and real-world evidence generation by correlating wearable-derived biomarkers with therapeutic outcomes, creating a closed-loop feedback system between clinical deployment and knowledge graph enrichment.
- **Measure for Success:** DHT data streams integrated and normalized within FAIR-compliant pipelines; wearable-derived endpoints incorporated into at least two active Digital Twin models; real-world evidence generation pipeline operational.

Roadmap Summary Matrix

Phase / Timeline	Stage	Key Deliverables	Success Criteria
Phase 1: Foundation (Months 1–6)	Stage I: Data Ingestion & Normalization	Unified data lake (200+ sources) MBSE-governed pipelines FAIR quality gateway	PhD accuracy at Big Data scale 100% FAIR compliance
	Stage II: Knowledge Extraction & Modeling	ARK property graph (30M+ nodes) 80+ relationship types	0.9 fidelity threshold Entity resolution validated
Phase 2: Acceleration (Months 6–12)	Stage III: Interface & Application Deployment	Target & Safety Dossiers (18 views) Pharma Science Oncology Platform Rare Hopes UX	Time-to-insight: months → minutes Patient advisory validation
	Patient Digital Twins & Context Engine	Digital Twin infrastructure AWS Bedrock Semantic Gateway Enterprise Prompt Library	Digital Twin concordance validated 100% provenance on graph answers
Phase 3: Evolution (Months 12–24)	Stage IV: Generative AI & Agentic Integration	Ask ARCH conversational AI Large Action Models (LAMs) Gaia recursive learning platform	>95% provenance accuracy Autonomous workflow execution
	DHT, Wearables & Digital Twin Maturation	Wearable data integration Living Digital Twin models Real-world evidence pipeline	DHT FAIR-compliant pipelines RWE generation operational

Semantic Governance Architecture: Ontology Lifecycle and Mapping Strategy

The ARCH knowledge platform depends on a rigorous semantic governance architecture that defines where ontologies are authored and maintained, how top-down semantic structures are reconciled with bottom-up data realities, and how that reconciliation is enforced consistently across the Neo4j graph topology. Without this governance layer, the “Architecture of Truth” collapses into another integration silo. This section specifies the three-tier ontology stack, the tooling and governance processes for each tier, and the mapping mechanisms that bind them into a coherent, continuously validated semantic fabric.

Three-Tier Ontology Stack

The ARCH semantic architecture operates as a three-tier ontology stack. Each tier has a distinct authorship model, governance cadence, and tooling environment, but all three are bound together through formal mapping assertions that guarantee semantic coherence from the highest level of abstraction down to the physical Neo4j node and edge labels.

Tier 1: Upper Ontology (Core 9)

The Core 9 ontologies—Process, Service, Asset, Product, Party, Location, Time, Event, and Issue—constitute the upper ontology and serve as the stable semantic ceiling for the entire ARCH ecosystem. These are not domain-specific vocabularies; they are abstract, enterprise-wide categories that define the fundamental types of things Pharma reasons about across all business domains. Their purpose is to guarantee that no matter how many domain ontologies or data sources are federated into ARCH, every concept can be traced upward to one of these nine root categories.

Authorship and Governance: The Core 9 is authored and maintained by the Enterprise Knowledge Accelerator (EKA) team in collaboration with the Knowledge Integration and Articulation (KIA) governance council. The upper ontology is modeled in OWL 2 and managed within a dedicated governance repository that is version-controlled independently from domain ontologies. Changes to the Core 9 follow a formal change-control process requiring cross-functional review from enterprise architecture, data governance, and domain science leadership. Release cadence is quarterly at most; the upper ontology is intentionally slow-moving to provide a stable foundation.

Tooling: The Core 9 is maintained in a formal ontology editor (Protégé or TopBraid-class tooling) that supports OWL 2 reasoning, SHACL constraint validation, and export to multiple serialization formats (RDF/XML, Turtle, JSON-LD). The canonical serialization is published to the SciBite CENTree ontology server as the authoritative reference vocabulary, making it accessible to all downstream consumers—including the domain ontology authors, the NER pipeline, and the Neo4j graph schema generator.

Tier 2: Domain Ontologies (Top-Down Semantic Layer)

Domain ontologies represent the therapeutic-area-specific and function-specific vocabularies that specialize the Core 9 for concrete R&D use. Examples include the Oncology Target Ontology (specializing Product and Asset for oncology molecules, biological targets, and biomarkers), the Clinical Safety Ontology (specializing Issue and Event for adverse events, toxicology endpoints, and regulatory findings), and the Immunology Pathway Ontology (specializing Process for immune signaling cascades and mechanism-of-action modeling). Each

domain ontology is formally aligned to the Core 9 through `rdfs:subClassOf` and `owl:equivalentClass` assertions, ensuring that every domain-specific concept inherits the enterprise-wide semantics of its parent category.

Authorship and Governance: Domain ontologies are authored collaboratively by domain scientists and knowledge engineers embedded within each therapeutic area. Authorship is democratized through SciBite CENTree, which provides a web-based vocabulary management interface that allows scientists to propose new terms, synonyms, and hierarchical relationships without requiring direct OWL editing skills. CENTree enforces structural constraints inherited from the Core 9—every new concept must be classified under an existing upper ontology branch before it can be promoted to production. Domain ontology governance follows a monthly review cadence managed by the KIA team, with automated validation checks confirming that all new concepts maintain valid subsumption paths to the Core 9.

Tooling: SciBite CENTree serves as the primary authoring environment for all domain ontologies. CENTree manages term lifecycle (draft, review, approved, deprecated), synonym sets, cross-references to external authorities (UniProt, ChEBI, MeSH, SNOMED CT, Gene Ontology), and hierarchical relationships. CENTree publishes versioned ontology releases that are consumed downstream by the NER pipeline (SciBite TERMite), the relationship extraction engine (Tellic), and the Neo4j graph schema generator. Every CENTree release triggers automated regression testing against the Core 9 constraint model to ensure structural integrity.

Tier 3: Bottom-Up Source Schemas and Neo4j Graph Topology

The bottom-up tier consists of the physical data schemas from 200+ internal and external sources, plus the Neo4j labeled property graph topology that materializes the ARK knowledge graph. These structures are inherently heterogeneous: source databases use proprietary column names, legacy codes, and inconsistent entity representations, while the Neo4j graph uses node labels, relationship types, and property keys that must conform to graph query performance requirements. The bottom-up tier is where the “disparate strings” live before they are resolved into “unified biological things.”

Source Schema Profiling: During Phase 1 ingestion, every source undergoes MBSE-governed schema profiling. This process inventories all tables, columns, data types, value distributions, and implicit entity references within each source. The profiling output is a machine-readable source schema model that captures the bottom-up reality of what exists in each database without imposing top-down assumptions.

Neo4j Graph Schema: The Neo4j graph topology is not a free-form graph. It is governed by a formal graph schema—a set of approved node labels, relationship types, and property key constraints—that is derived from the domain ontologies via a schema generation pipeline. Each domain ontology class maps to a Neo4j node label, each object property maps to a relationship type, and each data property maps to a node property key. This generation process is deterministic and version-controlled: when a domain ontology release introduces a new concept, the graph schema generator produces the corresponding Neo4j constraint definitions, which are reviewed, tested, and deployed through a standard CI/CD pipeline.

Top-Down to Bottom-Up Mapping Architecture

The critical architectural question is how the top-down ontology stack (Tiers 1 and 2) is reconciled with the bottom-up source schemas and Neo4j topology (Tier 3). ARCH addresses this through a three-stage mapping pipeline that operates as an automated, continuously validated bridge between semantic intent and physical data reality.

Stage A: Semantic Alignment Mapping (Source → Domain Ontology)

For each ingested source, a semantic alignment map is produced that binds every source schema element (column, code, entity reference) to its corresponding domain ontology concept. This mapping is a formal assertion: “Column X in Source Y represents instances of Concept Z in Domain Ontology D.” These mapping assertions are themselves managed as versioned, auditable artifacts stored in the ARCH metadata catalog.

The mapping process uses a hybrid approach. SciBite TERMite performs automated candidate matching by running NER against source metadata (column names, value samples, data dictionaries) to propose candidate ontology alignments. These candidates are then reviewed and validated by knowledge engineers using an AI-assisted mapping dashboard that presents confidence-ranked suggestions alongside the source schema profile. The MBSE traceability model guarantees completeness: every source schema element must be mapped, explicitly excluded, or flagged for remediation before the source is promoted to production ingestion. This prevents unmapped columns from silently entering the data lake as orphaned, semantically opaque artifacts.

Stage B: Graph Schema Projection (Domain Ontology → Neo4j Topology)

Once domain ontologies are published from CENtree, the graph schema projection pipeline translates the OWL/SKOS structures into the Neo4j labeled property graph topology. This projection follows deterministic transformation rules that ensure consistent, predictable graph structures.

- **Node Label Derivation:** Each domain ontology class that is flagged as “materializable” generates a corresponding Neo4j node label. The Core 9 upper classes are also materialized as supertype labels, enabling multi-label inheritance in the graph. For example, a node representing the molecule Pharma-951 carries both the domain label :TherapeuticCandidate and the upper ontology label :Product, allowing Cypher queries to operate at any level of abstraction.
- **Relationship Type Derivation:** Each object property in the domain ontology generates a Neo4j relationship type. The 80+ relationship types in ARK (inhibits, agonism, upregulates, ASSOCIATED_WITH_ADVERSE_EVENT, etc.) are all derived from formal ontology property definitions, not ad-hoc graph modeling decisions. Domain and range constraints from the ontology are enforced as Neo4j constraints, preventing semantically invalid edges from being created.
- **Property Key Derivation:** Data properties from the ontology generate Neo4j property keys with enforced data type constraints. Every node carries mandatory provenance properties—source identifier, ingestion timestamp, confidence score, and CENtree concept ID—ensuring full traceability from graph node back through the domain ontology to the originating source schema element.
- **Polymorphic Label Inheritance:** Neo4j’s multi-label capability is exploited to materialize the full ontological hierarchy on every node. A gene target node simultaneously carries labels from all three tiers: the domain-specific label (e.g., :KinaseTarget), the broader domain class (e.g., :BiologicalTarget), and the Core 9 supertype (e.g., :Asset). This enables the Context Engine to resolve queries at the appropriate level of specificity—a scientist querying “all kinase targets” retrieves a different scope than a strategist querying “all assets”—while both queries traverse the same underlying graph.

Stage C: Continuous Reconciliation and Drift Detection

Mapping is not a one-time activity. As source schemas evolve, as new data sources are onboarded, and as domain ontologies are refined, the alignment between tiers can drift. ARCH addresses this through continuous reconciliation mechanisms.

- Schema Drift Detection:** Automated monitors compare the current state of each source schema against its last-profiled baseline. When columns are added, removed, or changed in a source system, the drift detector flags the affected semantic alignment maps for review. This prevents silent data quality degradation where a source schema change causes downstream entities to be misclassified or lost.
- Ontology-Graph Consistency Validation:** After every domain ontology release from CENtree, an automated validation suite compares the published ontology against the live Neo4j graph schema. The validator checks for orphaned node labels (labels in the graph that no longer have ontology backing), missing materializations (new ontology concepts that have not yet been projected into the graph), constraint violations (edges that violate updated domain or range restrictions), and label hierarchy inconsistencies (nodes whose multi-label sets do not correctly reflect the current ontology subsumption hierarchy).
- Concept ID Anchoring:** Every node in the Neo4j graph carries a CENtree concept ID as a mandatory property. This concept ID is the immutable anchor that binds the physical graph node to its position in the ontological hierarchy. When a domain ontology concept is deprecated or merged, the concept ID anchoring ensures that all affected graph nodes are identified and remediated. The concept ID also serves as the join key between the graph and the FAIR metadata catalog, enabling end-to-end lineage tracing from Cypher query result back through the graph node, through the domain ontology, through the semantic alignment map, to the specific column and row in the originating source system.

Mapping Flow Summary

The complete mapping flow operates as a bidirectional governance loop. Top-down, the Core 9 constrains what domain ontologies may express; domain ontologies constrain what the Neo4j graph schema may contain; and the graph schema constrains what data may be materialized. Bottom-up, source schema profiling discovers what data actually exists; semantic alignment maps propose how that data maps to domain concepts; and unmapped or ambiguous elements trigger ontology extension requests back up through CENtree to the domain and, where necessary, upper ontology governance councils. This bidirectional loop is what prevents the ARCH knowledge graph from becoming either a rigid top-down imposition that ignores data reality or an ungoverned bottom-up accumulation that lacks semantic coherence.

Tier	Authorship	Tooling	Governance Cadence
Tier 1: Upper Ontology (Core 9)	EKA team + KIA governance council; cross-functional change control	OWL 2 editor (Protégé/TopBraid); published to CENtree as canonical reference	Quarterly (intentionally slow; stability mandate)

Tier 2: Domain Ontologies	Domain scientists + embedded knowledge engineers; democratized via CENTree	SciBite CENTree (web-based vocabulary management with Core 9 enforcement)	Monthly review with automated Core 9 subsumption validation
Tier 3: Source Schemas	Source system owners, profiled by MBSE-governed ingestion pipelines	Modak Nabu (automated profiling); TERMite (candidate alignment); AI mapping dashboard	Continuous (drift detection on every ingestion cycle)
Tier 3: Neo4j Graph Topology	Graph schema generated deterministically from domain ontology releases	Neo4j schema generator (CI/CD); SHACL/constraint validators	Triggered on every CENTree release; automated consistency validation

End-to-End Lineage: From Cypher Query to Source Row

The three-tier mapping architecture produces complete, auditable lineage for every piece of knowledge in the ARCH graph. When a scientist executes a Cypher query and receives a result—for example, “Pharma-951 inhibits KinaseTarget-X with confidence 0.94”—the lineage chain is fully traceable. The Neo4j node carries a CENTree concept ID that links to the domain ontology concept (e.g., Oncology Target Ontology: KinaseTarget-X). The domain ontology concept has a formal subsumption path to the Core 9 (KinaseTarget-X → BiologicalTarget → Asset). The node’s provenance properties identify the source (e.g., Tellic external extraction, PubMed article PMID:12345678). The semantic alignment map records which source schema element was mapped to this concept. And the MBSE traceability model confirms that the mapping was complete and validated. This lineage is what transforms ARCH from a data retrieval system into a GxP-compliant Architecture of Truth where every assertion can be defended under regulatory scrutiny.

Ontological Framework: Core 9 Mapping to ARCH

The ARCH platform is governed by a set of nine core ontologies—Process, Service, Asset, Product, Party, Location, Time, Event, and Issue—that provide the stable architectural lineage ensuring every data point remains traceable to an enterprise-wide Unifying Ontology. This upper-level semantic layer governs disparate, bottom-up data schemas while ensuring polymorphic meaning across different business domains.

Semantic Blueprint: Core 9 to Dossier Mapping

Core Ontology	ARCH Dossier Application	Strategic Value
Product	The therapeutic candidate or molecule being investigated	Ensures the Digital Twin of a molecule is consistent across all R&D silos (Lars Greiffenberg)
Asset	Internal genomics data, chemical DELs, and Tellic's 1B+ biomedical relationships	Provides clear inventory of Integrated Knowledge assets for GxP auditing (Michael Kaltschmidt)
Process	Drug discovery lifecycle from hypothesis generation to clinical trial execution	Maps LAM workflows to specific R&D milestones (Brian Martin)
Party	Scientists querying the graph, practitioners in the oncology portal, and patients	Drives User Enablement by tailoring UX to each party's vocabulary (Elfi Stach)
Location	Therapeutic areas (Oncology, Immunology) and global research sites	Harmonizes data originating from different global sites (Lake County vs. Ludwigshafen)
Time	Phased execution timeline and trial durations	Anchors the 24-month roadmap to measurable temporal milestones
Event	Data ingestion triggers, milestone achievements, and patient outcome observations	Tracks provenance by recording exactly when and how knowledge entered ARCH
Issue	Integration debt, AI hallucinations, and regulatory compliance risks	Uses the ontology to flag and track hallucination-free status for AI outputs
Service	AI-driven insight delivery, RAG frameworks, and the Gaia Platform	Transitions the ontology from a static database to an active research accelerator

Polymorphic Meaning and Integration Debt Resolution

The Core 9 mapping demonstrates that as a Product (molecule) undergoes transformation across different systems—from a string in a published paper to a thing in a chemical library—it maintains its identity via the Core Product Ontology. This recursive identity preservation is the key to solving integration debt: ARCH is not another silo but a system where every dossier view is a business-specific projection of the enterprise-wide core ontology.

Context Engine: Operational Demonstration

The following walkthrough illustrates how the Context Engine transforms a simple user query into a precise, ground-truth Cypher query, utilizing the 5 Pillars of Relevancy and Core 9 ontological mapping.

Step 1: Raw Input

User Query: "What is the latest on drug safety for our new oncology target?"

Step 2: Context Engine Analysis (5 Pillars)

Pillar	Extracted Ontological Metadata	Source	Core 9
Who	Senior Research Fellow (Lars Greiffenberg persona). Permissions: GxP / Internal R&D.	Party	Party
What	Target: Pharma-951 (mapped from "new oncology target"). Concept: Adverse Events & Toxicology.	Product / Asset	Product / Asset
Where	Domain: Oncology R&D. Site: Lake County / ARCH Production Graph.	Location	Location
When	Scope: Post-January 2026 (filtered for "latest" clinical trial uploads).	Time / Event	Time / Event
Why	Intent: Target/Safety Dossier authoring for regulatory submission.	Process / Issue	Process / Issue

Step 3: Semantic Gateway (Cypher Generation)

The Enterprise Prompt Library uses the enriched metadata to generate a precise Cypher query:

```
// Generated by Context Engine for Persona: R&D_Scientist
MATCH (p:Product {name: 'Pharma-951'})-[:HAS_TARGET]->(t:BiologicalTarget)
MATCH (t)-[r:ASSOCIATED_WITH_ADVERSE_EVENT]-(ae:Issue)
WHERE r.timestamp > '2026-01-01' AND r.source = 'Internal_Clinical_Trial'
RETURN p.name, t.name, ae.description, r.severity, r.timestamp ORDER BY
r.timestamp DESC LIMIT 5
```

Step 4: Verified Articulation

Instead of a generic LLM response, the UI renders a Safety Dossier view with grounded, traceable results:

For Target Pharma-951: 3 new Grade 2 neutropenia events were recorded in the ARCH Hub since your last login (Feb 2026). These events correlate with Tellic's external findings for similar protein kinase inhibitors.

Every element of this response is traceable: the Who (persona and permissions), What (specific molecule and adverse event type), Where (production graph and oncology domain), When (post-January 2026 filter), and Why (dossier authoring intent) are all auditable, satisfying GxP compliance requirements.

Stakeholder Value Delivery

The Patient

Knowledge integration accelerates the delivery of first-in-kind medicines, which now represent 75% of Pharma's pipeline. ARCH has already demonstrated value in drug repurposing, identifying potential therapies for complex diseases and COVID-19 by flagging existing molecules with high-fidelity potential. Patient Digital Twins further personalize therapeutic approaches by simulating individual patient responses before clinical intervention, while the Rare Hopes UX ensures that patients with rare diseases gain access to graph-derived insights that would otherwise remain locked in research literature.

The Practitioner and Scientist

The framework empowers scientists through the Target and Safety Dossiers, which aggregate 18 distinct views of a target into one interface. Researchers receive a comprehensive view of pathways, variants, and phenotypes in minutes, pivoting from data gathering to innovative experimentation. The Pharma Science Oncology Platform extends this value to external practitioners, delivering curated research updates directly to oncologists. During Phase 3, wearable-derived DHT endpoints provide objective, real-world patient outcome data that complements traditional clinical measures.

The Pharma Enterprise

The ARCH platform catalyzes the immunology encore, bolstering expansion into neuroscience and obesity as primary growth drivers. Operationally, the platform prevents duplication of expensive datasets and ensures supply resilience across future product cycles. The unified knowledge layer provides quantifiable ROI through reduced trial redundancy, compressed discovery timelines, and lower late-stage attrition rates.

Cultural Transformation: The Pharma Convergence Journal

Value realization is anchored in cultural change. The Pharma Convergence Journal—a quarterly, internal, peer-reviewed publication—showcases real-world use cases where scientists build their own products atop the ARCH ecosystem. This transformational commitment ensures the platform becomes a centerpiece of Pharma's converged scientific culture.

Data Governance & Quality Standards

FAIR Principles as Foundational Mandate

All data entering the ARCH ecosystem must satisfy strict adherence to FAIR Principles across all four dimensions. These are not aspirational guidelines but unyielding operational mandates enforced through automated quality gating.

- **Findability:** Every data asset receives persistent, unique identifiers and is registered in a searchable metadata catalog. Rich descriptive metadata enables discovery across the 200+ connected sources.
- **Accessibility:** Data is retrievable through standardized, open protocols with clearly defined authentication and authorization mechanisms. Access policies are persona-aware, governed by the Context Engine's Party ontology.
- **Interoperability:** All data uses formal, shared, broadly applicable vocabularies managed through SciBite CENtree. Ontological mapping ensures that concepts are semantically interoperable across therapeutic domains and organizational boundaries.
- **Reusability:** Data assets carry clear provenance metadata, usage licensing, and domain-relevant community standards. Every asset is documented for secondary use, enabling risk-based data democratization across the R&D enterprise.

MBSE as Core Ingestion Methodology

Model-Based Systems Engineering serves as the primary methodology governing all data ingestion and integration workflows within ARCH. Rather than ad-hoc ETL development, every integration pipeline is designed, validated, and documented as a formal systems engineering model. MBSE guarantees zero data loss during migration from legacy systems by providing complete traceability from source schema to target ontology, structural validation of mapping completeness before execution, automated regression testing against known data integrity benchmarks, and prevention of integration debt through rigorous architectural governance. This approach ensures that the ARCH data foundation is structurally sound from inception, avoiding the accumulation of technical debt that plagues conventional data lake implementations.

Service Level Agreements (SLAs)

Unyielding SLAs govern data quality, availability, and freshness across all ARCH data assets. These SLAs are monitored continuously through automated dashboards and are integrated into the FAIR quality gateway. Non-compliant data assets are quarantined until remediation is complete, preventing erroneous or stale data from contaminating the knowledge graph.

Strategic Risk Mitigation

The ARCH platform serves as a primary mitigation strategy for the four core strategic risks facing Pharma:

Debt & Capital Risk

By increasing R&D efficiency and preventing late-stage clinical failure, ARCH maximizes the return on R&D investment to balance high debt levels. Digital Twin simulations further reduce capital risk by optimizing trial designs and reducing redundant experiments before capital is committed.

Execution Risk

Pharma is investing \$380 million in new AI-powered manufacturing facilities in Chicago. ARCH acts as the pipeline feeder for these sites, ensuring they are utilized for high-quality, validated medicines. The 24-month phased roadmap with explicit milestones provides continuous visibility into execution progress.

Regulatory Pressure

A traceable knowledge graph provides the evidence and provenance required to satisfy intensified pricing and regulatory scrutiny. The Context Engine's full auditability—capturing Who, What, Where, When, and Why for every query and response—delivers GxP-compliant documentation automatically.

Margin Stability

By shortening discovery timelines and driving the immunology encore in oncology and obesity, the framework protects margins against competitive pressures. The integration of DHT and wearable data in Phase 3 generates real-world evidence that supports value-based pricing and market access strategies.

Leadership Engagement Strategy

The ARCH solution architecture directly addresses the strategic priorities of key stakeholders across the organization:

- **Bob Martin:** The Context Engine demonstration proves that the Semantic Gateway is not a prompt wrapper but a systematic architecture guaranteeing RAG Grounding by forcing the LLM to speak the language of the graph (Cypher) rather than guessing. The LAM evolution in Phase 3 maps action model workflows directly to R&D milestones.
- **Mark Jones:** The FAIR-compliant, MBSE-governed ingestion pipeline ensures GxP-readiness. Every query is traceable to the Who, What, and When, ensuring that data articulation is compliant and auditable. The Core 9 mapping demonstrates that ARCH resolves integration debt through recursive identity preservation, not silo construction.
- **Leslie Griffin:** The Product ontology ensures the Digital Twin of a molecule is consistent across all R&D silos. Phase 2's Patient Digital Twins extend this concept to patient-level simulation, creating a comprehensive twin ecosystem spanning molecules and patients.
- **Sherry North:** The Party ontology drives User Enablement by tailoring UX to each persona's vocabulary. The Rare Hopes UX and Pharma Science Oncology Platform demonstrate concrete platform extensions reaching patients and practitioners beyond internal R&D.

This Solution Architecture serves as a blueprint for building an agentic framework to transform the biopharmaceutical enterprise, unlocking the information that makes cures possible.